**Table S3.** Exploratory X1/X2: group summaries of the per-cell Δ for every analysis, arm, comparator and group.

| Arm | Comparator | Group | Cells | Mean Δ (95% interval) | Cells with Δ < 0 |
|---|---|---|---|---|---|
| X1: eraser (LEACE) refitted on the 158-case analysis region | | | | | |
| FM-P−C | step-6 FM-P−C | P1b-FM | 48 | −0.033 (−0.042 to −0.023) | 34 |
| FM-P−C | step-6 FM-P−C | P2-FM | 12 | −0.002 (−0.007 to 0.005) | 6 |
| FM-P−C | step-6 FM-P−C | slice | 12 | −0.010 (−0.022 to 0.002) | 9 |
| FM-P−C | step-6 FM-P | P1b-FM | 48 | 0.561 (0.549 to 0.573) | 0 |
| FM-P−C | step-6 FM-P | P2-FM | 12 | −0.007 (−0.013 to −0.001) | 7 |
| FM-P−C | step-6 FM-P | slice | 12 | −0.003 (−0.014 to 0.009) | 6 |
| FM-P−C | step-6 NV | P1b-FM | 48 | −1.250 (−1.277 to −1.224) | 48 |
| FM-P−C | step-6 NV | P2-FM | 12 | −0.024 (−0.032 to −0.013) | 7 |
| FM-P−C | step-6 NV | slice | 12 | −0.378 (−0.392 to −0.363) | 11 |
| FM-S−C | step-6 FM-S−C | P1b-FM | 48 | 0.023 (0.014 to 0.033) | 18 |
| FM-S−C | step-6 FM-S−C | P2-FM | 12 | 0.000 (−0.005 to 0.006) | 7 |
| FM-S−C | step-6 FM-S−C | slice | 12 | 0.000 (−0.012 to 0.012) | 7 |
| FM-S−C | step-6 FM-S | P1b-FM | 48 | 0.343 (0.333 to 0.353) | 1 |
| FM-S−C | step-6 FM-S | P2-FM | 12 | −0.004 (−0.009 to 0.002) | 7 |
| FM-S−C | step-6 FM-S | slice | 12 | 0.006 (−0.006 to 0.019) | 4 |
| FM-S−C | step-6 NV | P1b-FM | 48 | −1.247 (−1.270 to −1.222) | 46 |
| FM-S−C | step-6 NV | P2-FM | 12 | −0.019 (−0.028 to −0.009) | 7 |
| FM-S−C | step-6 NV | slice | 12 | −0.384 (−0.400 to −0.369) | 11 |
| REF−C | step-6 REF−C | P1b-FM | 48 | 0.004 (−0.006 to 0.012) | 18 |
| REF−C | step-6 REF−C | P2-FM | 12 | 0.002 (−0.003 to 0.008) | 5 |
| REF−C | step-6 REF−C | slice | 12 | 0.000 (−0.012 to 0.011) | 5 |
| REF−C | step-6 REF | P1b-FM | 48 | 0.496 (0.484 to 0.508) | 1 |
| REF−C | step-6 REF | P2-FM | 12 | −0.006 (−0.013 to −0.001) | 6 |
| REF−C | step-6 REF | slice | 12 | 0.004 (−0.008 to 0.016) | 5 |
| REF−C | step-6 NV | P1b-FM | 48 | −1.280 (−1.307 to −1.256) | 47 |
| REF−C | step-6 NV | P2-FM | 12 | −0.022 (−0.030 to −0.012) | 7 |
| REF−C | step-6 NV | slice | 12 | −0.406 (−0.421 to −0.392) | 11 |
| X2: linear-only nuisance learners | | | | | |
| FM-P−C | step-6 FM-P−C | P1b-FM | 48 | 0.011 (0.002 to 0.020) | 22 |
| FM-P−C | step-6 FM-P−C | P2-FM | 12 | 0.001 (−0.004 to 0.008) | 7 |
| FM-P−C | step-6 FM-P−C | slice | 12 | −0.003 (−0.014 to 0.008) | 7 |

| Arm | Comparator | Group | Cells | Mean Δ (95% interval) | Cells with Δ < 0 |
|---|---|---|---|---|---|
| FM-P−C | step-6 FM-P | P1b-FM | 48 | 0.605 (0.593 to 0.617) | 0 |
| FM-P−C | step-6 FM-P | P2-FM | 12 | −0.004 (−0.010 to 0.002) | 7 |
| FM-P−C | step-6 FM-P | slice | 12 | 0.004 (−0.008 to 0.016) | 3 |
| FM-P−C | step-6 NV | P1b-FM | 48 | −1.206 (−1.233 to −1.179) | 47 |
| FM-P−C | step-6 NV | P2-FM | 12 | −0.021 (−0.030 to −0.012) | 7 |
| FM-P−C | step-6 NV | slice | 12 | −0.371 (−0.387 to −0.355) | 11 |
| FM-S−C | step-6 FM-S−C | P1b-FM | 48 | −0.003 (−0.012 to 0.006) | 27 |
| FM-S−C | step-6 FM-S−C | P2-FM | 12 | −0.000 (−0.006 to 0.005) | 6 |
| FM-S−C | step-6 FM-S−C | slice | 12 | −0.001 (−0.013 to 0.011) | 6 |
| FM-S−C | step-6 FM-S | P1b-FM | 48 | 0.317 (0.306 to 0.328) | 0 |
| FM-S−C | step-6 FM-S | P2-FM | 12 | −0.004 (−0.010 to 0.002) | 7 |
| FM-S−C | step-6 FM-S | slice | 12 | 0.004 (−0.008 to 0.017) | 3 |
| FM-S−C | step-6 NV | P1b-FM | 48 | −1.273 (−1.299 to −1.248) | 47 |
| FM-S−C | step-6 NV | P2-FM | 12 | −0.020 (−0.029 to −0.010) | 7 |
| FM-S−C | step-6 NV | slice | 12 | −0.386 (−0.401 to −0.370) | 11 |
| REF−C | step-6 REF−C | P1b-FM | 48 | −0.000 (−0.009 to 0.008) | 21 |
| REF−C | step-6 REF−C | P2-FM | 12 | −0.001 (−0.006 to 0.005) | 7 |
| REF−C | step-6 REF−C | slice | 12 | 0.002 (−0.010 to 0.013) | 7 |
| REF−C | step-6 REF | P1b-FM | 48 | 0.492 (0.480 to 0.504) | 0 |
| REF−C | step-6 REF | P2-FM | 12 | −0.009 (−0.015 to −0.003) | 7 |
| REF−C | step-6 REF | slice | 12 | 0.005 (−0.006 to 0.018) | 5 |
| REF−C | step-6 NV | P1b-FM | 48 | −1.285 (−1.310 to −1.258) | 47 |
| REF−C | step-6 NV | P2-FM | 12 | −0.024 (−0.033 to −0.015) | 7 |
| REF−C | step-6 NV | slice | 12 | −0.405 (−0.420 to −0.389) | 11 |

Exploratory (outside the frozen protocol; plan registered on OSF, osf.io/5c627); no verdict. Δ = (|bias| X arm − |bias| step-6 arm) / mean truth per cell, always X arm − step-6 arm (negative: the X arm has the smaller |bias|); mean Δ = unweighted mean over the group's cells; 95% interval: paired bootstrap percentile interval (2,000 resamples of the 200 synthetic replicates within each cell; cells fixed); simulation uncertainty, not real-data sampling uncertainty.

Groups: P1b-FM, the 48 W = 0 cells with P = 5 or 30; P2-FM, the 12 cells with K = 1, P = 0, α > 0; slice, the 12 main-text slice cells (W = 30, α = 0.5, γ = 1, $\varphi_A$ = 0); cell 15 belongs to P2-FM and the slice. −C, linearly site-removed; NV, no video.

Ceiling (fact base §8): exploratory; no verdict; cannot restore P1b-FM/P2-FM; nonlinear site information remains after X1; X2 changes do not show that erasure caused a difference.